

Model selection for Markov random fields on graphs under a mixing condition

XVII CLAPEM

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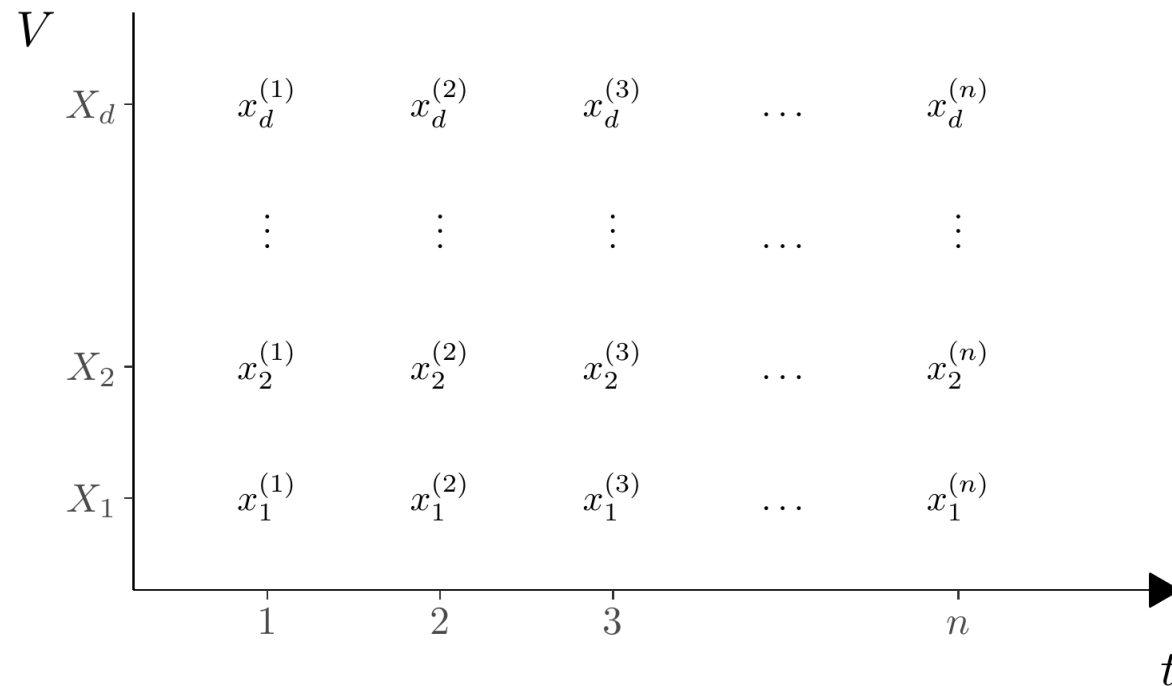
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Agenda

- Motivation
- Mathematical Framework
- Estimation Method and Theoretical Results
- Applications to real data
- Discussion and Future Work

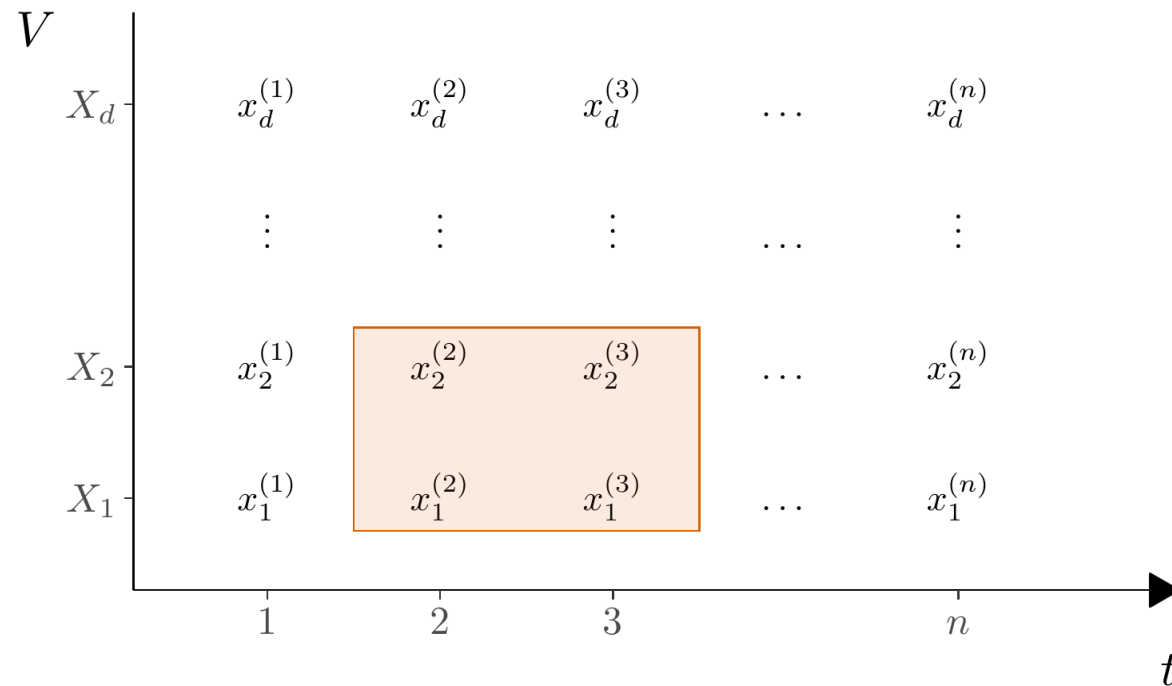
Vector-Valued Stochastic Processes

- $\mathbf{X}^{(\mathbf{i})} = (\mathbf{X}_1^{(\mathbf{i})}, \mathbf{X}_2^{(\mathbf{i})}, \dots, \mathbf{X}_d^{(\mathbf{i})})$.
- $X_v^{(i)} \in A$, A a finite alphabet, for $v \in \{1, 2, \dots, d\}$.
- Stationary and ergodic process $\mathbf{X} = \{\mathbf{X}^{(\mathbf{i})} : -\infty < \mathbf{i} < \infty\}$ in $((A^d)^{\mathbb{N}}, \mathcal{F}, \mathbb{P})$.
- We assume the process $\mathbf{X}$ has an underlying graph $G^* = (V, E^*)$.



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The IID case

- **Classic Problem:** In the IID scenario, we have the standard model selection problem for discrete graphical models or Markov random fields on graphs.
- **Established Literature:** Extensive research exists for this setting, including works by Lauritzen (1996), Koller and Friedman (2009), and Leonardi et al. (2024).
- **Common Approaches:**
 - **Structure Estimation:** Addressed via standard logistic regression (for binary models), distance-based methods, or penalized pseudo-likelihood.
 - **Neighborhood-Based Estimation:** Many methods involve estimating individual neighborhoods for each vertex and then combining them to form the final graph.

Why go beyond the IID assumption?

Most traditional graphical model techniques assume that observations $\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots$ are Independent and Identically Distributed (IID).

However, in real-world scenarios, the independence assumption is often too restrictive:

- **EEG Time Series:** Neural dependencies evolve over time.
- **River Stream Flow:** Successive observations exhibit hydrological dependence.
- **Financial Markets:** Daily indices show significant autocorrelation and mixing properties.

Our Proposal: A global model selection criterion that ensures consistency even under mixing conditions.

Objectives of the research

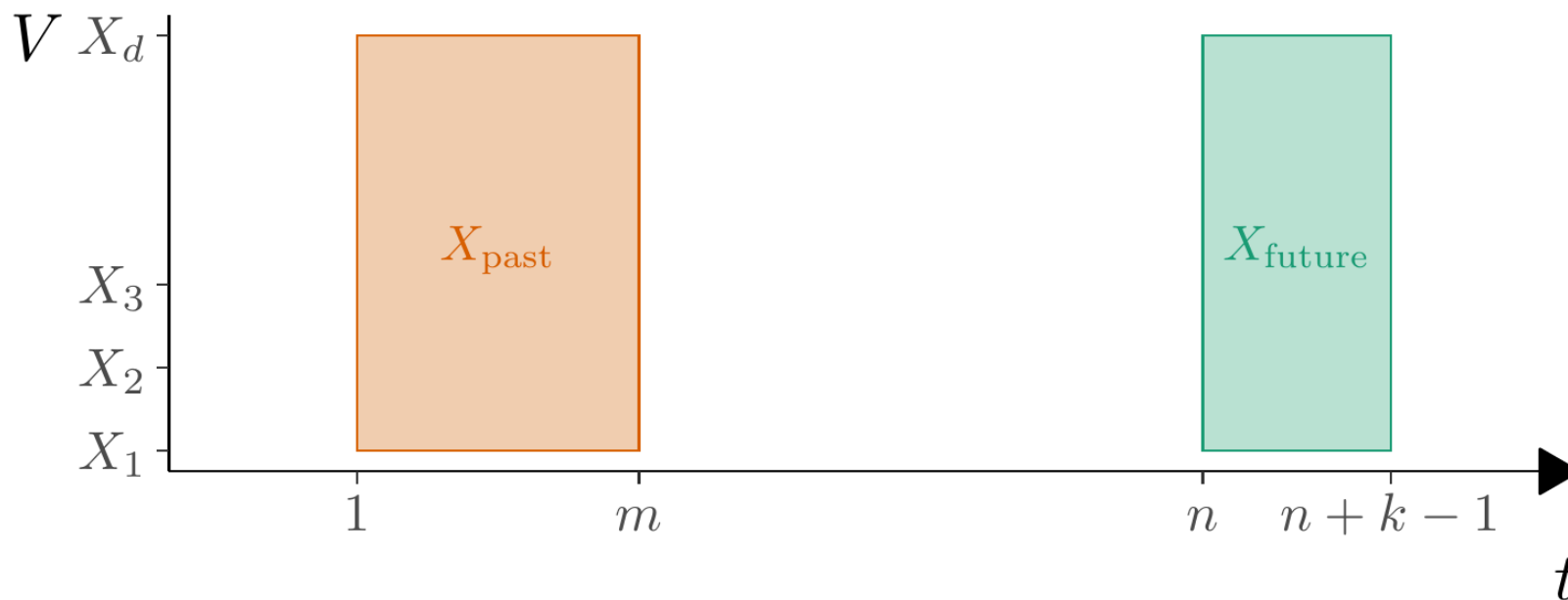
- **Proposal:** Method to estimate the graph of conditional dependencies for multivariate stochastic processes with mixing conditions.
- **Aim:** combine and generalize previous works,
 - Leonardi et al. (2021): method assumes decomposition into subvectors with immediate neighbor dependencies,
 - Leonardi et al. (2023): estimator of neighborhood of each vertex for iid data.
- **Proposed solution:** penalized pseudo-likelihood criterion for entire graph estimation for multivariate processes with mixing conditions.
- **Key advantages:**
 - Handles non-iid data (mixing condition),
 - Provides a global estimation approach.

Mixing Condition

- Let $X^{(i:j)}$ denote the sequence of vectors $X^{(i)}, X^{(i+1)}, \dots, X^{(j)}$.

The process satisfies a **mixing condition** with rate $\psi(\ell)$ if the dependence between the past ($X^{(1:m)}$) and the future ($X^{(n:n+k-1)}$) decays as the distance $\ell = n - m$ increases:

$$\left| \mathbb{P}(X_{future} | X_{past}) - \mathbb{P}(X_{future}) \right| \leq \psi(n - m) \mathbb{P}(X_{future}) \quad (1)$$



Empirical Probabilities

As the stationary distribution of the process π is unknown, we must estimate it from the data

Assume we observe a sample of size n of the process, denoted by $\{x^{(i)} : i = 1, \dots, n\}$. Then, for any $W \subset V$ and any $a_W \in A^W$ ($\pi(a_W) = \mathbb{P}(X_W = a_W)$),

$$\widehat{\pi}(a_W) = \frac{N(a_W)}{n}.$$

where $N(a_W)$ denotes the number of times the configuration a_W appears in the sample. If $\widehat{\pi}(a_W) > 0$,

$$\widehat{\pi}(a_U | a_W) = \frac{\widehat{\pi}(a_{U \cup W})}{\widehat{\pi}(a_W)},$$

for two disjoint subsets $W, U \subset V$ and configurations $a_W \in A^W, a_U \in A^U$.

Empirical Probabilities

Given a graph $G = (V, E)$, define

$$G(v) = \{u \in V : (u, v) \in E\},$$

for $v \in V$, the set of neighbors of v in graph G .

Then we can compute

$$\widehat{\pi}(a_v | a_{G(v)}) = \frac{\widehat{\pi}(a_{\{v\} \cup G(v)})}{\widehat{\pi}(a_{G(v)})}.$$

Graph Estimator

Given any graph G and a sample of the process, we define the pseudo-likelihood function by

$$\widehat{L}(G) = \prod_{i=1}^n \prod_{v \in V} \widehat{\pi}(x_v^{(i)} | x_{G(v)}^{(i)}) . \quad (2)$$

Applying the logarithm we can write expression above as

$$\log \widehat{L}(G) = \sum_{v \in V} \sum_{(a_v \in A)} \sum_{a_{G(v)} \in A^{|G(v)|}} N(a_v, a_{G(v)}) \log \widehat{\pi}(a_v | a_{G(v)}) \quad (3)$$

The regularized graph estimator is given by

$$\widehat{G} = \arg \max_G \left\{ \log \widehat{L}(G) - \lambda_n \sum_{v \in V} |A|^{|G(v)|} \right\} .$$

Consistency Theorem

Theorem (Leonardi and Severino, 2025): Assume the process $\{X^{(i)} : i \in \mathbb{Z}\}$ satisfies the mixing condition presented before with $\psi(\ell) = O(1/\ell^{1+\epsilon})$ for some $\epsilon > 0$. Then, by taking $\lambda_n = c \log n$, we have that

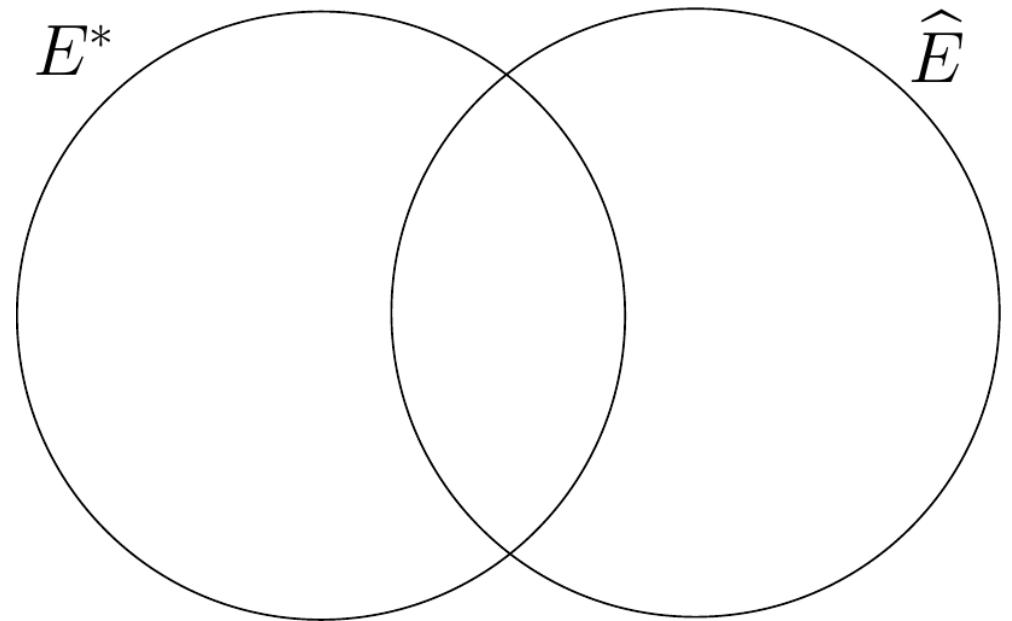
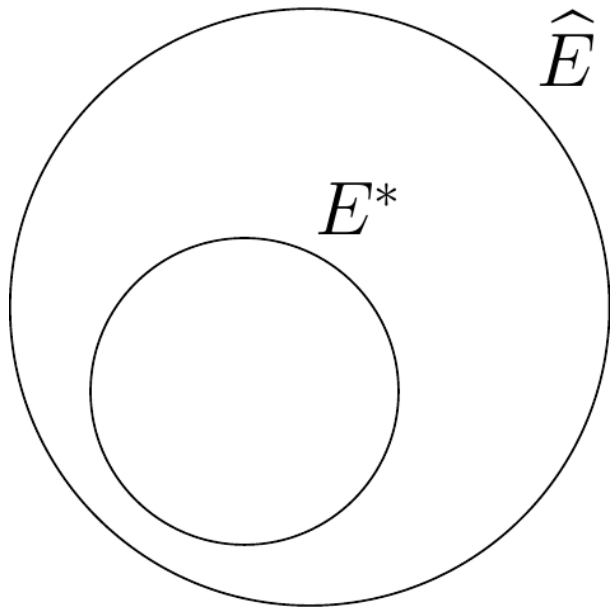
$$\widehat{G} = \arg \max_G \left\{ \log \widehat{L}(G) - \lambda_n \sum_{v \in V} |A|^{G(v)} \right\}$$

satisfies $\widehat{G} = G^*$ eventually almost surely as $n \rightarrow \infty$.

Intuitive Overview of Theorem Proof

The proof of this consistency is based on results about the rate of convergence of empirical probabilities.

Consider $\{\hat{G} \neq G^*\} = \{G^* \subsetneq \hat{G}\} \cup \{G^* \not\subset \hat{G}\}$.



We prove that, eventually almost surely as $n \rightarrow \infty$, neither of the cases above can happen, which implies that $\hat{G} = G^*$.

Algorithms for estimation

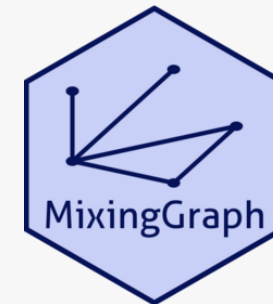
The objective is to find the optimal graph by maximizing the regularized log-pseudo-likelihood function:

$$H(G) = \log \widehat{L}(G) - \lambda_n \sum_{v \in V} |A|^{G(v)}.$$

Implemented Search Strategies

- **Exact Algorithm:** Performs an exhaustive search over the set of all possible simple graphs $\mathcal{G}$. Computational complexity: $2^{d(d-1)/2}$.
- **Stepwise (Forward/Backward):** Heuristic approach that adds or removes edges based on local improvements of $H(G)$.
- **Simulated Annealing:** A stochastic global search metaheuristic used to approximate the estimator in high-dimensional spaces.

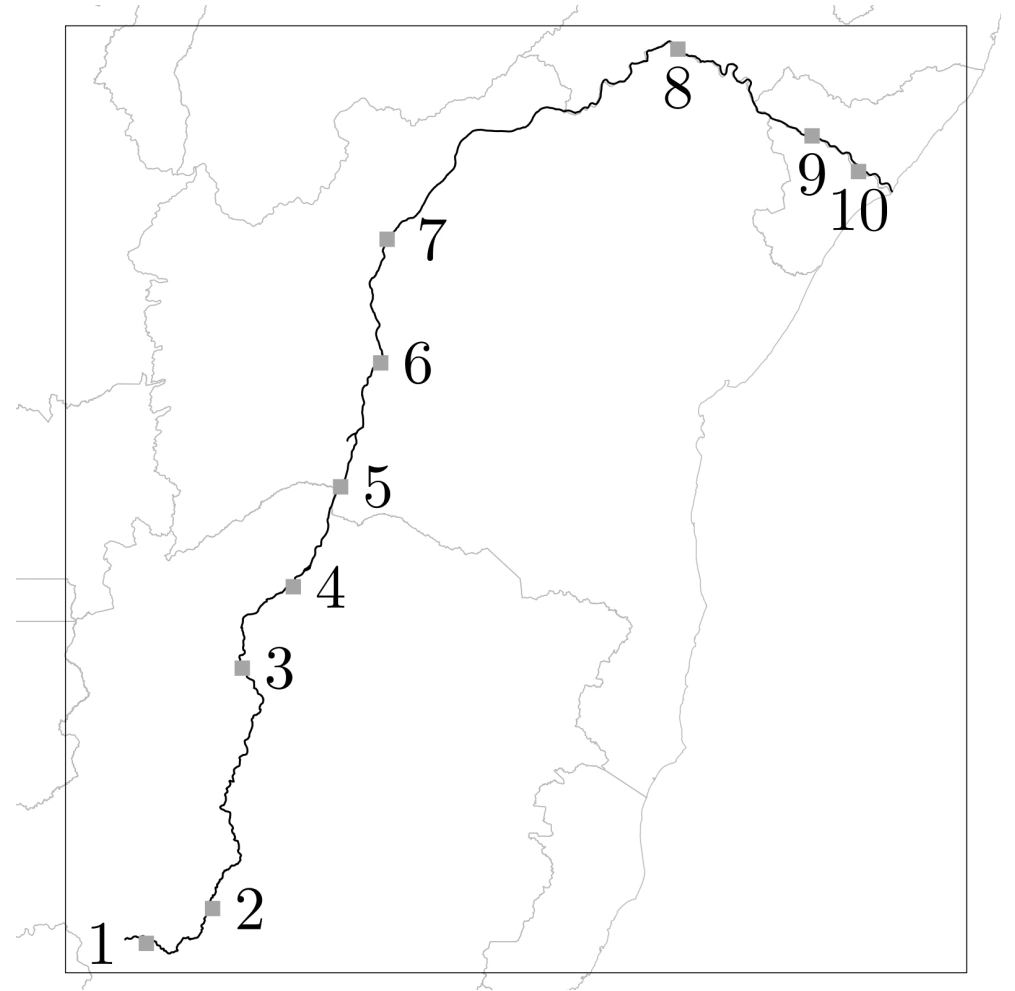
The MixingGraph Package



Comprehensive tools for estimating the graph $\hat{G}$, executing search algorithms, and ensuring reproducible results.

Application to real data

São Francisco River Data

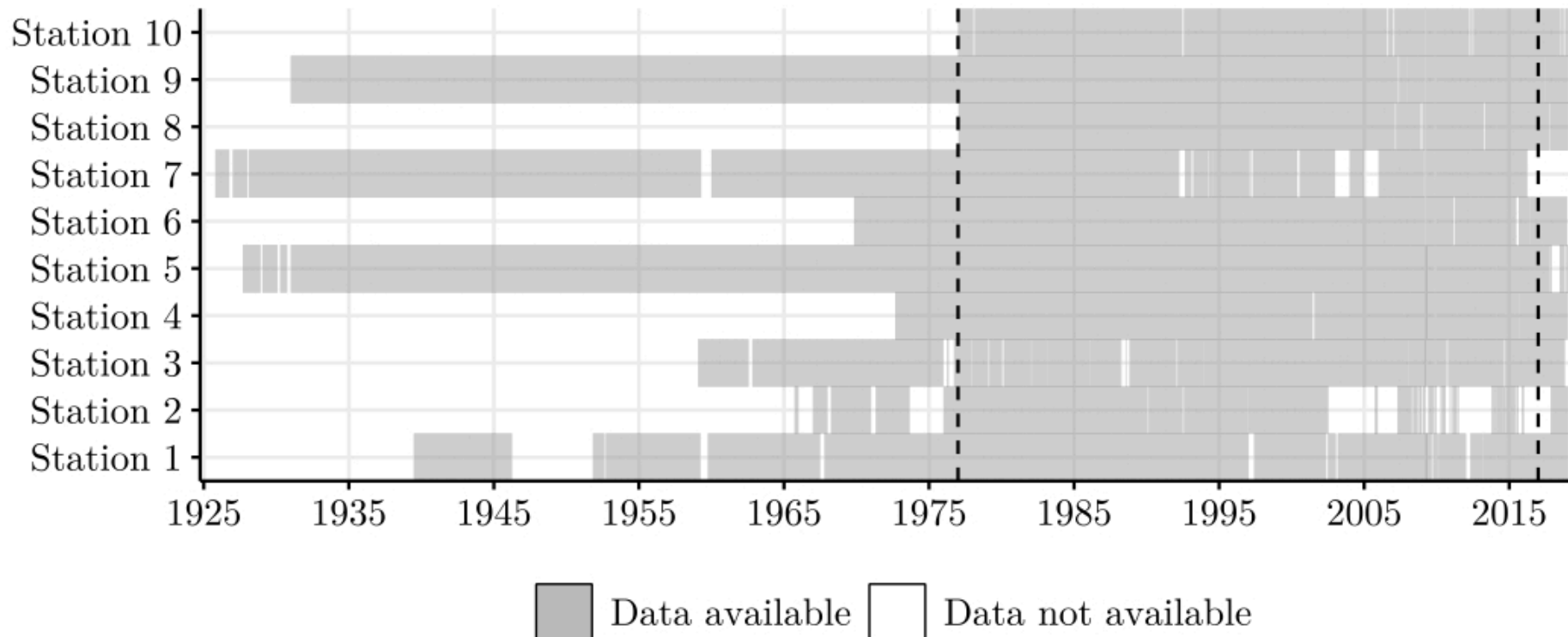


São Francisco River Data

- Water volume measured at d stations along the river's course.
- Stations denoted as X_u , where $u = 1, \dots, d = 10$.
- Vector $\mathbf{X}$ observed at discrete time intervals (10-day mean).
- Each observation denoted as $\mathbf{X}^{(i)} = (X_1^{(i)}, \dots, X_{10}^{(i)})$.
- Process $\mathbf{X}^n = \{\mathbf{X}^{(i)} : 1 \leq i \leq n\}$.

São Francisco River Data

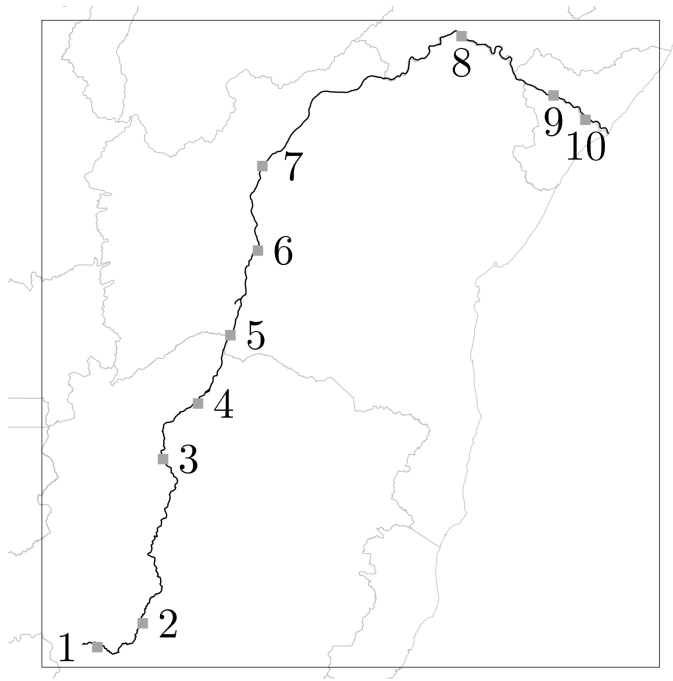
- **Data availability:** daily measurements from October 1st, 1924, to February 28th, 2019.
- **Data considered:** from January 1977 to December 2016.
- Total of 1042 observations.
- **Discretization** of the data into five levels based on quantiles.



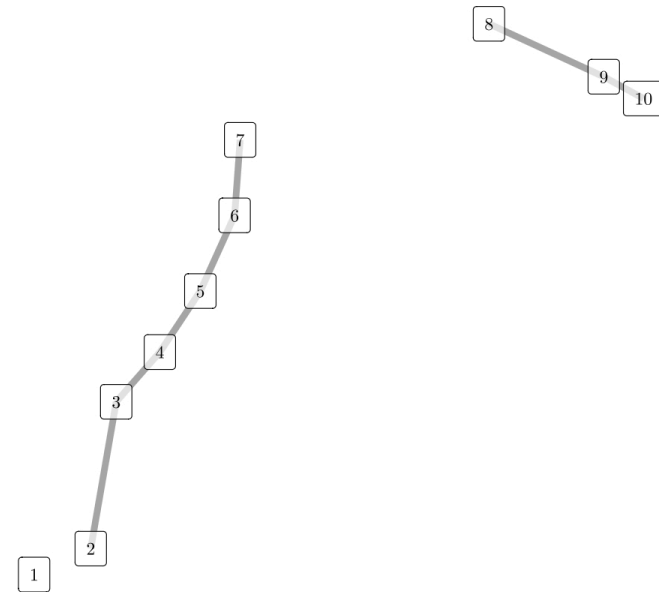
São Francisco River: Structural Results

We applied the **Forward Stepwise Algorithm** with **5-fold cross-validation** to select the optimal penalty c .

Physical Network



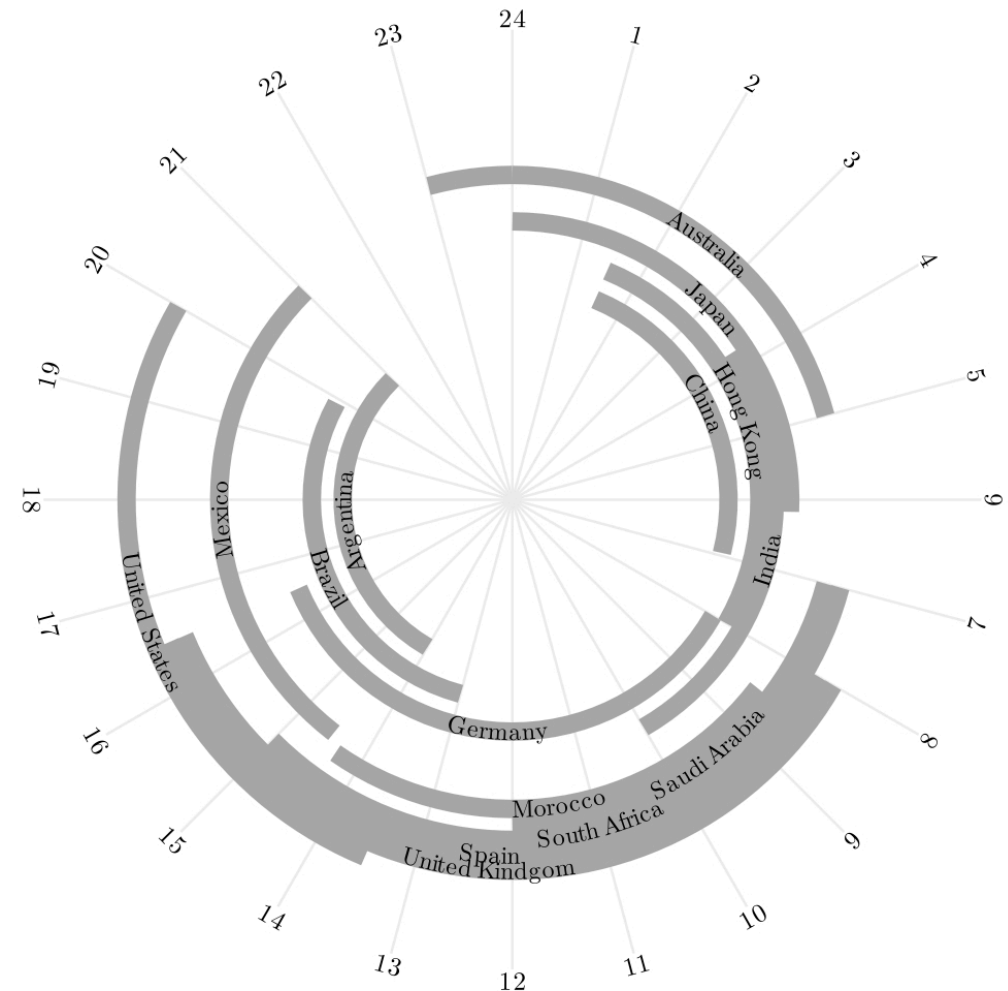
Estimated Graph $\hat{G}$



- The estimated edges accurately recover the hydrological flow and physical connectivity of the river.
- Results are consistent with independent block identification methods (Leonardi et al, 2021)

Stock Exchange Data

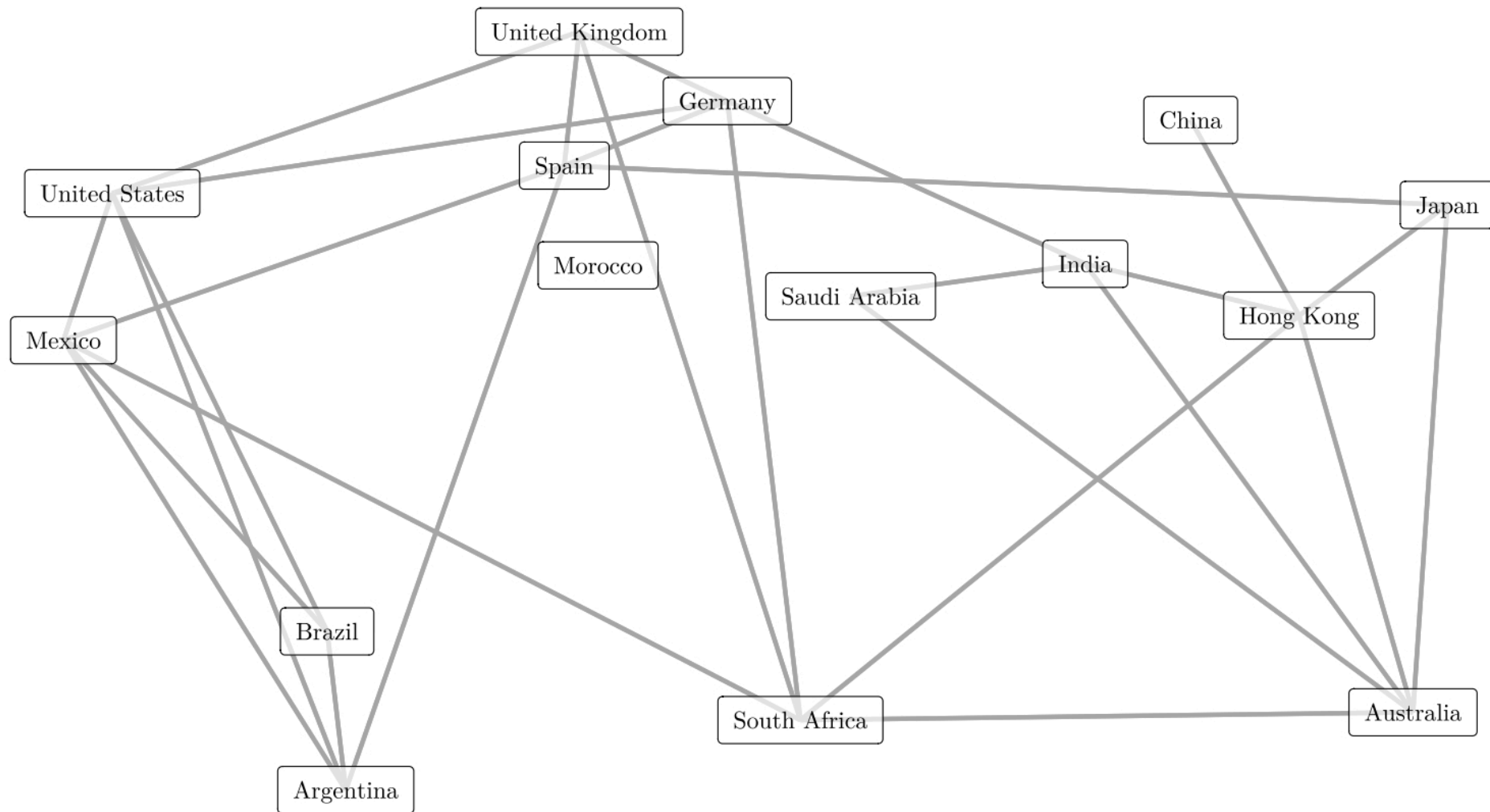
- Relation between stock exchanges' performance reflect global market interconnectedness and economic trends.
- Impacts investment strategies, risk assessment, and portfolio diversification.
- **Challenge:** Variations in stock exchange operating hours due to different time zones.
- Complex analysis required to account for global market fluctuations.



Stock Exchange Data

- **Objective:** analyse the Relation between stock exchanges' performance.
- **Discretization:** daily indicator of an increase in the index rate.
- Data from May 18th, 2010 to September 20th, 2023.
- Adjustments made to address missing data and holiday schedules.
- Final dataset comprises 2,654 rows for analysis.
- Forward stepwise algorithm and a 5-fold cross-validation approach.

Stock Exchange Data



Estimated graph, considering the penalizing constant value chosen by cross-validation.

Discussion & Future Work

- **Summary:**

- Generalization of structure recovery for multivariate processes under **mixing conditions**.
- Development of a **global estimation criterion** based on penalized log-pseudo-likelihood.
- Proved **strong consistency** and derived convergence rates for the estimator $\hat{G}$.
- Implementation of exhaustive and heuristic search algorithms (Stepwise, Simulated Annealing).
- Validation through extensive simulations and applications to **hydrological** and **financial** data.

- **Future Directions:**

- Generalization to **infinite vertex sets** and unbounded degree estimators.
- Extension to **continuous-valued** multivariate stochastic processes.

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¡Gracias!
Obrigado!
Thank you!